

Edge AI – Principles and Practices

Module – Edge AI in the real world

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Module Objectives

- This module explores some Edge AI applications in the real world and their design considerations.

Common use cases

□ Why Edge AI?

- ❖ 50% of the Earth's land is relatively untouched by human development.
- ❖ It is impractical/too costly to provide reliable/constant power in those areas.
- ❖ Data is huge in modern applications.
- ❖ Sending huge data for remote processing is not the best solution (or too expensive)
- ❖ Local processing is the choice (Edge AI)

Greenfield projects

- ❑ Project site/condition/environment/infrastructure is new/untouched/primitive/uncultivated.
- ❑ Pros: a blank sheet, freedom to choose latest/new technologies
- ❑ Cons: risky (may not be successful), costly

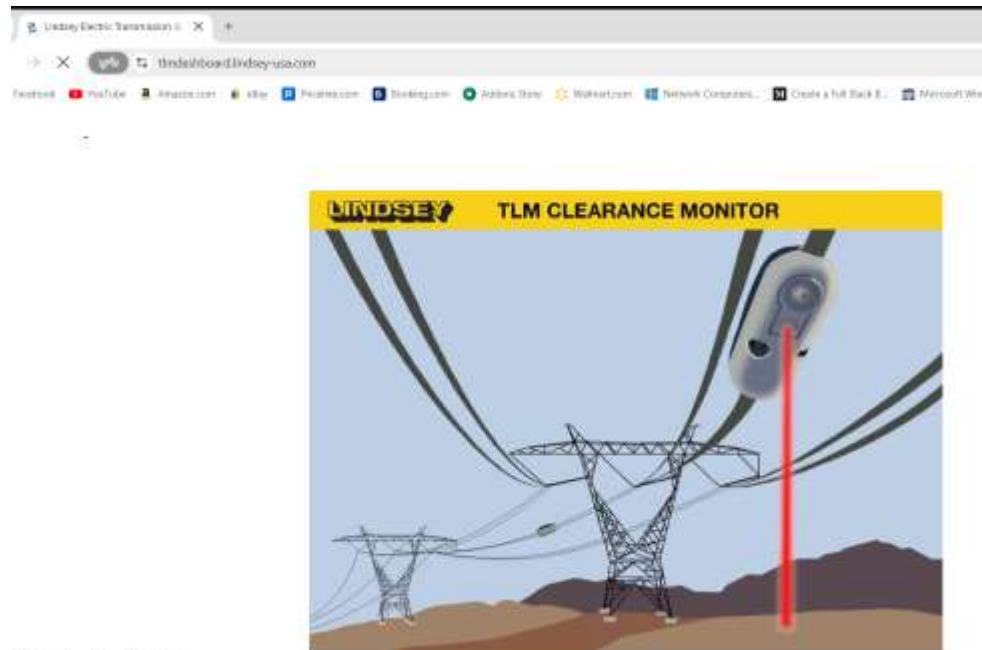
Brownfield projects

- ❑ Project site/condition/environment/infrastructure exists.
- ❑ Pros: less expensive, lower risk, faster development
- ❑ Cons: constraints

Real-world products

□ Power line monitoring

- ❖ Design experience (from the class instructor)



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Real-world products

❑ Wearable

- ❖ Design experience (from the class instructor)



Real-world products

❑ Wearable

- ❖ Design experience (from the class instructor)



Real-world products

❑ Wearable

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Responsible design and AI ethics

- ❑ AI has great impact on humanity.
- ❑ Therefore, AI Ethics is an important topic.

Responsible design and AI ethics

❑ Bias

- ❖ Favor some and prejudice others

❑ Discrimination

- ❖ Treat individuals or groups differently

❑ Fairness

- ❖ Hard to define (what is fair?)
- ❖ Fairness (equity, equality, inclusiveness)

Closing Thoughts

- ❑ Lots of real-world applications
 - ❖ Lots of business opportunities
- ❑ How do you prepare yourself?
 - ❖ Don't stock up too many technologies
 - ❖ Do have a "silo project" in which you executed tasks from the top to the bottom.
- ❑ Ethics - Have a professional bottom line (don't cross it)